Lecture Notes:
Commutative Algebra
&
Introduction to Algebraic Geometry

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(Based on the notes taken by Friedrich Günther and Niklas Müller in Summer Term 2019)

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Chapter 1

Rings and Ideals

In this chapter, we introduce and study some basic notions and results concerning rings and ideals. Note that some of them were already presented in the courses "Lineare Algebra I" and "Algebra", but are included here for completeness and clarity.

Throughout these notes, all rings are assumed to be *commutative with identity*, unless explicitly stated otherwise. In particular, any nontrivial ring $R \neq \{0\}$ contains at least two different elements: 0 and 1.

Definition 1.1. Let R and S be rings. A *ring homomorphism* $\varphi\colon R \to S$ is a map fulfilling the following properties:

(a) $\varphi(x+y) = \varphi(x) + \varphi(y)$ for all $x, y \in R$,

(b) $\varphi(xy) = \varphi(x)\varphi(y)$ for all $x, y \in R$,

(c) $\varphi(1_R) = 1_S$.

Note that property (c) does not hold automatically and hence must be required.

Example 1.2. Let R and S be nontrivial rings. The following maps between rings fulfill (a) and (b), but not (c).

(1) The zero map $R \to S,\ x \mapsto 0$.

(2) The map $R \to R \times R,\ x \mapsto (x, 0)$.

(3) The map $\mathbb{Z}/6\mathbb{Z} \to \mathbb{Z}/6\mathbb{Z},\ \overline{n} \mapsto \overline{3} \cdot \overline{n}$.

Definition 1.3. Let R be a ring. A subset $S \subseteq R$ is a *subring* of R if it is a ring with respect to the operations induced by those in R and moreover $1_R \in S$.

Remark 1.4. Let R be a ring and let $S \subseteq R$ be a subset.

(a) The subset S is a subring if and only if it is nonempty, closed under subtraction and under multiplication, and $1_R \in S$.

(b) If S is a subring, then the inclusion map $S \to R$, $x \mapsto x$, is a ring homomorphism.

Remark 1.5. In the definition of a subring $S \subseteq R$, the condition $1_R \in S$ is not automatically satisfied and must be imposed. Indeed, a subset of a ring may itself form a ring under the induced operations without containing the identity element of the larger ring. For example, let R be a nontrivial ring and consider

$$S := \{(x, 0) \mid x \in R\} \subseteq R \times R.$$

The subset S is a ring under the operations inherited from $R \times R$: indeed, S is nonempty, closed under subtraction and multiplication, and $(1, 0)$ is the multiplicative identity of S. However, the identity element of $R \times R$ is $(1, 1)$ and since R is nontrivial we have $(1, 0) \neq (1, 1)$. Since $(1, 1) \notin S$, the ring S is not a subring of $R \times R$.

Definition 1.6. Let R be a ring and let $a, b \in R$. We say that b *divides* a, or that b *is a divisor of* a, and write $b \mid a$, if there exists an element $c \in R$ such that $a = bc$.

Definition 1.7. Let R be a ring and let $r \in R$.

(a) The element r is called a *unit* if it is invertible with respect to multiplication, i.e., there exists $s \in R$ such that $rs = 1$. We denote by R^* the set of all units in R. Notice that $(R^*, \cdot)$ is an abelian group.

(b) The element r is called a *zero divisor* if there exists $s \in R \setminus \{0\}$ such that $rs = 0$. A nontrivial ring whose only zero divisor is 0 is called an *integral domain*.

(c) The element r is called *nilpotent* if there exists $n \in \mathbb{N}_{>0}$ such that $r^n = 0$.

(d) Assume now that $r \neq 0$ and $r \notin R^*$.

(i) The element r is called *prime* if for all $a, b \in R$

$$r \mid ab \quad \Longrightarrow \quad r \mid a \quad \text{or} \quad r \mid b.$$

(ii) The element r is called *irreducible* if for all $a, b \in R$

$$r = ab \quad \Longrightarrow \quad a \in R^* \quad \text{or} \quad b \in R^*.$$

Remark 1.8.

(a) In a nontrivial ring any nilpotent element is a zero divisor. However, the converse is not true in general. For instance, in the ring $\mathbb{Z}/6\mathbb{Z}$ the element $\overline{2}$ is a zero divisor, but it is not nilpotent.

(b) In general, the notions of prime and irreducible elements are independent: a prime element need not be irreducible, and an irreducible element need not be prime. For instance, in the ring $\mathbb{Z} \times \mathbb{Z}$ the element $(1,0)$ is prime, but not irreducible since $(1,0) = (1,0) \cdot (1,0)$. Also, in the ring $\mathbb{Z}[i\sqrt{5}] := \{a + bi\sqrt{5} \mid a, b \in \mathbb{Z}\}$ the element 2 is irreducible, but not prime. Thus, without additional assumptions on the ring, there is no implication between the two concepts. However, in special classes of rings the situation improves:

(i) In an integral domain, every prime element is irreducible.

(ii) In a unique factorization domain (UFD), the two notions coincide.

1.1 Ideals: Examples and Operations

Definition 1.9. Let $(R, +, \cdot)$ be a ring. A subset $I \subseteq R$ is called an *ideal* if:

(a) $(I, +)$ is a subgroup of $(R, +)$, and

(b) for all $r \in R$, for all $a \in I$, it holds $ra \in I$.

We give now some examples of ideals and define operations on them.

Example 1.10. Let R be a ring.

(a) The subsets $\{0\}$ and R are ideal and are called *trivial ideals*.

(b) Given two ideals $I, J \subseteq R$, we define

$$\begin{aligned} I + J &:= \Big\{ i + j \mid i \in I,\ j \in J \Big\}, \\ IJ &:= \left\{ \sum_{k=1}^{n} i_k j_k \mid n \in \mathbb{N},\ i_k \in I,\ j_k \in J \right\}, \\ I \cap J &:= \Big\{ a \mid a \in I \text{ and } a \in J \Big\}. \end{aligned}$$

All three of them are ideals and it holds $IJ \subseteq I \cap J$. Notice that

$$I \cup J := \Big\{ a \mid a \in I \text{ or } a \in J \Big\}$$

is an ideal if and only if $I \subseteq J$ or $J \subseteq I$.

(c) Let $\{I_\alpha\}_{\alpha \in \mathcal{A}}$ be a family of ideals in R. We define

$$\begin{aligned} \sum_{\alpha \in \mathcal{A}} I_\alpha &:= \left\{ \sum_{\alpha \in \mathcal{A}} i_\alpha \;\middle|\; i_\alpha \in I_\alpha \text{ for all } \alpha \in \mathcal{A} \text{ and } i_\alpha \neq 0 \text{ only for finitely many } \alpha \right\}, \\ \bigcap_{\alpha \in \mathcal{A}} I_\alpha &:= \Big\{ a \;\Big|\; a \in I_\alpha \text{ for all } \alpha \in \mathcal{A} \Big\}. \end{aligned}$$

Both of them are ideals. Now, assume that $\{I_\alpha\}_{\alpha\in\mathcal{A}}$ is a chain of ideals, i.e., for any $\alpha, \beta \in \mathcal{A}$, either $I_\alpha \subseteq I_\beta$ or $I_\beta \subseteq I_\alpha$. Then,

$$\bigcup_{\alpha\in\mathcal{A}} I_\alpha := \Big\{a \;\Big|\; a \in I_\alpha \text{ for some } \alpha \in \mathcal{A}\Big\}$$

is also an ideal.

(d) Let $a \in R$. We define

$$(a) := \{\lambda a \mid \lambda \in R\}.$$

This is an ideal called *the ideal generated by* a; we denote it by aR as well.

(e) Let $I \subseteq R$ be an ideal. We say that I is *principal* if there exists $a \in R$ such that $I = (a)$.

(f) Let $a_1, \ldots, a_n \in R$. We define

$$(a_1, \ldots, a_n) := a_1R + \ldots + a_nR = \left\{ \sum_{i=1}^{n} \lambda_i a_i \;\Big|\; \lambda_i \in R \right\}.$$

This is called *the ideal generated by* $a_1, \ldots, a_n$.

(g) Let $\{a_i\}_{i\in\mathcal{A}}$ be a subset of R. We define

$$(a_i)_{i\in\mathcal{A}} := \sum_{i\in\mathcal{A}} a_i R_i.$$

This is called *the ideal generated by the subset* $\{a_i\}_{i\in\mathcal{A}}$.

(h) Let $I \subseteq R$ be an ideal. We say that the subset $\{a_i \mid i \in \mathcal{A}\} \subseteq I$ is a *system of generators for* I if $I = (a_i)_{i\in\mathcal{A}}$. The ideal I is said to be *finitely generated* if there exists a finite system of generators for I.

Remark 1.11.

(a) Let R be a ring and let $I \subseteq R$ be an ideal. If $1_R \in I$, then $I = R$. In particular, a proper ideal $I \subsetneq R$ is not a subring of R.

(b) A proper ideal $I \subsetneq R$ of a ring R might be itself a ring. In this case, $1_I \neq 1_R$ is an idempotent element of R. For instance, the ring $S \subseteq R \times R$ given in Remark 1.5 is an ideal in $R \times R$. A natural question then arises: does there exist an example of a subset of a ring that is itself a ring but neither a subring nor an ideal of the ambient ring? The answer is positive. Indeed, the same ring S can be regarded as a subset of the polynomial ring $(R \times R)[x]$, but it is neither a subring nor an ideal of $(R \times R)[x]$.

1.2 Prime Ideals and Maximal Ideals

Definition 1.12. Let R be a ring and let $I \subsetneq R$ be a proper ideal.

(a) The ideal I is called *prime* if for any $a, b \in I$

$$ab \in I \implies a \in I \text{ or } b \in I.$$

(b) The ideal I is called *maximal* if for any ideal J

$$I \subseteq J \subseteq R \implies J = I \text{ or } J = R.$$

Proposition 1.13. *Let R be a ring and let $I \subsetneq R$ be a proper ideal.*

(i) The ideal I is prime if and only if R/I is an integral domain.

(ii) The ideal I is maximal if and only if R/I is a field.

Corollary 1.14. *In a ring any maximal ideal is prime.*

The converse does not hold true in general.

Example 1.15. Let R be a ring and let $I \subsetneq R$ be a proper ideal.

(a) If R is an integral domain that is not a field, then $I = (0)$ is prime, but not maximal.

(b) In $R = \mathbb{Z}[x, y]$, the ideal $I = (x)$ is prime, but not maximal since

$$(x) \subsetneq (x, y) \subsetneq \mathbb{Z}[x, y].$$

Proposition 1.16. *Let R be a ring and let $I \subseteq R$ be an ideal. There is a one-to-one inclusion-preserving correspondence between the ideals of R containing I and the ideals of R/I. Under this correspondence, prime (resp. maximal) ideals correspond to prime (resp. maximal) ideals.*

Hint: Let $\pi \colon R \to R/I$ be the canonical projection. If J is an ideal in R/I, then $\pi^{-1}(J)$ is an ideal in R containing I.

Proof. Exercise. □

Proposition 1.17. *(i) Every nontrivial ring $R \neq 0$ has a maximal ideal.*

(ii) Let R be a ring and let $I \subsetneq R$ be a proper ideal. Then there exists in R a maximal ideal containing I.

Proof. (i) Let $\mathcal{I} := \{I' \subsetneq R \mid I' \text{ is an ideal of } R\}$. Since $(0) \in \mathcal{I}$, the set $\mathcal{I}$ ordered by inclusion is a nonempty partially ordered set (poset for short). Let $\{I_\alpha\}_{\alpha\in\mathcal{A}}$ be a chain in $\mathcal{I}$, i.e., a totally ordered family of proper ideals in R. Since $\bigcup_{\alpha\in\mathcal{A}} I_\alpha$ is a proper ideal, every totally ordered subset of $\mathcal{I}$ has an upper bound in $\mathcal{I}$. Hence, by Zorn's Lemma, there exists a maximal element for the poset $(\mathcal{I}, \subseteq)$, i.e., a maximal ideal in R.

(ii) Apply (i) to the quotient ring R/I and then the one-to-one inclusion-preserving correspondence given by Proposition 1.16. Alternatively, argue as in (i) setting

$$\mathcal{I} := \{I \subseteq I' \subsetneq R \mid I' \text{ is an ideal of } R\}.$$

□

Corollary 1.18. *Every non-unit of a ring R is contained in a maximal ideal.*

Proof. Let $r \in R \setminus R^*$. Then the principal ideal (r) is a proper ideal of R. Now we apply Proposition 1.17 (ii) and we are done. □

Proposition 1.19. *Let R be a ring.*

(i) Let $\mathfrak{p}_1, \ldots, \mathfrak{p}_n \subsetneq R$ be prime ideals and let $I \subseteq R$ be an ideal such that $I \subseteq \bigcup_{i=1}^n \mathfrak{p}_i$. Then $I \subseteq \mathfrak{p}_i$ for some $i \in \{1, \ldots, n\}$.

(ii) Let $I_1, \ldots, I_n \subseteq R$ be ideals and let $\mathfrak{p} \subsetneq R$ be a prime ideal such that $\mathfrak{p} \supseteq \bigcap_{j=1}^n I_j$. Then $\mathfrak{p} \supseteq I_j$ for some $j \in \{1, \ldots, n\}$. Moreover, if $\mathfrak{p} = \bigcap_{j=1}^n I_j$, then $\mathfrak{p} = I_j$ for some $j \in \{1, \ldots, n\}$.

Proof. (i) The statement will be shown by induction on n in the form: If $I \not\subseteq \mathfrak{p}_i$ for $1 \le i \le n$, then $I \not\subseteq \bigcup_{i=1}^n \mathfrak{p}$.

For $n = 1$ the statement is trivial. Let now $n \ge 2$ and assume the statement is true for $n - 1$. Then for each $1 \le i \le n$ by the inductive hypothesis there exists $x_i \in I$ such that $x_i \notin \mathfrak{p}_j$ for all $j \ne i$. If for some $1 \le i \le n$ it holds $x_i \notin \mathfrak{p}_i$, then $x_i \notin \mathfrak{p}_j$ for all $1 \le j \le n$, thus we are done. Otherwise, it holds $x_i \in \mathfrak{p}_i$ for all $1 \le i \le n$. Consider the element

$$y := \sum_{i=1}^n x_1 x_2 \ldots x_{i-1} x_{i+1} \ldots x_n.$$

Note that $y \in I$ and $y \notin \mathfrak{p}_i$ for all $1 \le i \le n$. Indeed, assume by contradiction that $y \in \mathfrak{p}_i$ for some i. Then $x_1 x_2 \ldots x_{i-1} x_{i+1} \ldots x_n \in \mathfrak{p}_i$ since each of the other summands is a multiple of $x_i \in \mathfrak{p}_i$ and hence in $\mathfrak{p}_i$. But then, since $\mathfrak{p}_i$ is prime, we would have $x_j \in \mathfrak{p}_i$ for some $j \ne i$, a contradiction. Hence $y \notin \bigcup_{i=1}^n \mathfrak{p}_i$ and we are done.

(ii) Assume that $\mathfrak{p} \not\supseteq I_j$ for all $1 \leq j \leq n$. Then there exists $x_j \in I_j$ but $x_j \notin \mathfrak{p}$ for each $1 \leq j \leq n$. Thus

$$\prod_{j=1}^{n} x_j \in \prod_{j=1}^{n} I_j \subseteq \bigcap_{j=1}^{n} I_j.$$

But, since $\mathfrak{p}$ is prime, $\prod_{j=1}^{n} x_j \notin \mathfrak{p}$ and thus $\mathfrak{p} \not\supseteq \bigcap_{j=1}^{n} I_j$.

Now, if $\mathfrak{p} = \bigcap_{j=1}^{n} I_j$, then in particular we have $\mathfrak{p} \supseteq \bigcap_{j=1}^{n} I_j$ and therefore there is some $j_0 \in \{1, \ldots, n\}$ such that $I_{j_0} \subseteq \mathfrak{p}$. On the other hand, $\mathfrak{p} \subseteq I_j$ for $1 \leq j \leq n$, hence $\mathfrak{p} = I_{j_0}$.

□

1.3 Nilradical, Jacobson Radical and Local Rings

Proposition/Definition 1.20. *Let R be a ring and let $I \subseteq R$ be an ideal. The set*

$$\sqrt{I} := \{a \in R \mid a^n \in I \text{ for some } n \in \mathbb{N}_{>0}\}$$

is an ideal called the radical of I.

Proof. Let $r \in R$ and let $a \in \sqrt{I}$. Then $a^n \in I$ for some $n \in \mathbb{N}_{>0}$ and so $(ra)^n = r^n a^n \in I$ since I is an ideal. Now let $x, y \in \sqrt{I}$. Then there exist $n, m \in \mathbb{N}_{>0}$ such that $x^n, y^m \in I$. By the binomial formula we get

$$(x+y)^{m+n-1} = \sum_{k=0}^{m+n-1} \binom{m+n-1}{k} \underbrace{x^k y^{m+n-1-k}}_{\in I} \quad ,$$

hence, $(x+y)^{m+n-1} \in I$, which implies $x + y \in \sqrt{I}$. Thus, we have shown that $\sqrt{I}$ is indeed an ideal.

□

Proposition 1.21. *Let R be a ring. Let I and J be two ideals in R. Then:*

(i) $\sqrt{I} \supseteq I$,

(ii) $\sqrt{\sqrt{I}} = \sqrt{I}$,

(iii) $\sqrt{IJ} = \sqrt{I \cap J} = \sqrt{I} \cap \sqrt{J}$.

(iv) $\sqrt{I} = R \iff I = R$.

Definition 1.22. Let R be a ring. The *nilradical of R* is defined as

$$N_R := \sqrt{(0)} = \{a \in R \mid a^n = 0 \text{ for some } n \in \mathbb{N}_{>0}\}.$$

In other words, the set of nilpotent elements in R is indeed an ideal called the nilradical of R.

Proposition 1.23. *Let R be a ring. Then in the quotient ring R/N_R the only nilpotent element is* 0.

Proof. Let $[x] \in R/N_R$ be nilpotent. This means there is $k \in \mathbb{N}$ such that $[x]^k = [x^k] = 0$ in R/N_R, hence $x^k \in N_R$. Because $x^k \in N_R$, there is $l \in \mathbb{N}$ such that $(x^k)^l = x^{kl} = 0$ in R, and thus $x \in N_R$. Because $x \in N_R$, $[x] = 0$ in R/N_R. □

The following theorem describes the nilradical of a ring and, more in general, the radical of any ideal.

Theorem 1.24. *Let R be a ring and let $I \subseteq R$ be an ideal. Then:*

(i) The nilradical of R is the intersection of all prime ideals of R, i.e.,

$$N_R = \bigcap_{\substack{P \subsetneq R \\ prime}} P.$$

(ii) The radical of I is the intersection of all prime ideals containing I, i.e.,

$$\sqrt{I} = \bigcap_{\substack{I \subseteq P \subsetneq R \\ prime}} P.$$

Proof. (i) "$\subseteq$": Let $P' \subsetneq R$ be a prime ideal in R. If $x \in N_R$, there is $k \in \mathbb{N}$ such that $x^k = 0 \in P'$. Since P' is prime, it holds $x \in P'$, and thus $N \subseteq \bigcap_{\substack{P \subsetneq R \\ \text{prime}}} P$.

"$\supseteq$": Let now $x \in R \setminus N_R$ and let $\mathcal{S}_x$ be the set of proper ideals $I \subsetneq R$ with the property that if $n \in \mathbb{N}_{>0}$, then $x^n \notin I$. Since $(0) \in \mathcal{S}_x$, the set $\mathcal{S}_x$ is a nonempty poset (with respect to inclusion). As in the proof of Proposition 1.17 (i), by Zorn's Lemma $\mathcal{S}_x$ has a maximal element, say P'.

We show now that P' is a prime ideal in R. To see this, let $a, b \in R \setminus P'$. Then the ideals $(a) + P'$ and $(b) + P'$ are strictly larger than P' (with respect to inclusion) and thus are not in $\mathcal{S}_x$. Since $(a) + P'$, $(b) + P' \notin \mathcal{S}_x$, there is $m \in \mathbb{N}_{>0}$ such that $x^m \in (a) + P'$ and there is $k \in \mathbb{N}_{>0}$ such that $x^k \in (b) + P'$. Thus $x^m x^k = x^{m+k} \in [(a) + P'][(b) + P'] \subseteq (ab) + P'$. Since $x^{m+k} \in (ab) + P'$, the ideal $(ab) + P' \notin \mathcal{S}_x$, and thus $(ab) + P' \supsetneq P'$, which implies $ab \notin P'$. Finally, we have found a prime ideal P' such that $P' \in \mathcal{S}_x$. By definition of $\mathcal{S}_x$, the ideal P' does not contain any power of x. In particular, $x \notin P'$ and hence $x \notin \bigcap_{\substack{P \subsetneq R \\ \text{prime}}} P$.

(ii) Let $\pi \colon R \to R/I$ be the canonical projection. Note that $\pi^{-1}(N_{R/I}) = \sqrt{I}$. Hence, we first apply (i) to the ring R/I and then use the correspondence between prime ideals in R containing I and prime ideals in R/I given by Proposition 1.16.

□

Definition 1.25. Let R be a ring. The *Jacobson radical* J_R *of* R is definded as the intersection of all maximal ideals, i.e.,

$$J_R := \bigcap_{\substack{\mathfrak{m} \subsetneq R \\ \text{maximal}}} \mathfrak{m}.$$

Proposition 1.26. *Let R be a ring. The Jacobson radical J_R admits the following description*

$$J_R = \{x \in R \mid 1 - xy \in R^* \text{ for all } y \in R\}.$$

Proof. "$\subseteq$": Let $x \in J_R$ and assume that $1 - xy$ is not a unit for some $y \in R$. Then by Proposition 1.17 (ii) the element $1 - xy$ is contained in some maximal ideal $\mathfrak{m}$ of R and by definition of Jacobson radical, $x \in \mathfrak{m}$ as well. But then $1 \in \mathfrak{m}$, which is a contradiction.

"$\supseteq$": If $x \notin J_R$, then $x \notin \mathfrak{m}$ for some maximal ideal $\mathfrak{m}$ in R. Hence, $(x)+\mathfrak{m} \supsetneq \mathfrak{m}$ and thus $(x)+\mathfrak{m} = (1)$, i.e., there are elements $y \in R$ and $z \in \mathfrak{m}$ such that $xy+z = 1$. In other words, the element $1 - xy = z \in \mathfrak{m}$ and therefore $1 - xy$ is not a unit. □

Definition 1.27. A ring R is called *local* if it contains only one maximal ideal $\mathfrak{m}$. Then the quotient $R/\mathfrak{m}$ is a field, called the *residue field* of R. In literature, a local ring is often denoted by $(R, \mathfrak{m})$ or $(R, \mathfrak{m}, k)$, where $k := R/\mathfrak{m}$.

Proposition 1.28. *(i) Let R be a ring and let $\mathfrak{m} \subsetneq R$ be an ideal such that every element in $R \setminus \mathfrak{m}$ is a unit. Then R is a local ring and $\mathfrak{m}$ is its unique maximal ideal.*

(ii) Let R be a ring and let $\mathfrak{m} \subsetneq R$ be a maximal ideal such that every element in the set $1 + \mathfrak{m} := \{1 + x \mid x \in \mathfrak{m}\}$ is a unit. Then R is local and $\mathfrak{m}$ is its unique maximal ideal.

Proof. Exercise. □

1.4 Chinese Remainer Theorem

Definition 1.29. Let R be a ring and let $I, J \subseteq R$ be ideals. Then I and J are called *coprime* (or *comaximal*) if $I + J = R$.

Remark 1.30. Let R be a ring and I, J be ideals in R.

(i) If the ideals I, J are coprime, it holds $IJ = I \cap J$.

(ii) Two ideals $I, J \subseteq R$ are coprime if and only if there exist $i \in I$, $j \in J$ such that $i + j = 1$.

Proposition 1.31 (Chinese Remainder Theorem)**.** *Let R be a ring and $I_1, \dots, I_n \subseteq R$ be ideals. We define a ring homomorphism by*

$$\varphi\colon R \longrightarrow \prod_{j=1}^{n} (R/I_j), \qquad x \longmapsto (x + I_1, \dots, x + I_n).$$

(i) If I_k and I_l are coprime for $k \neq l$, then $\prod_{j=1}^{n} I_j = \bigcap_{j=1}^{n} I_j$.

(ii) φ is surjective if and only if I_k and I_l are coprime for $k \neq l$.

(iii) φ is injective if and only if $\bigcap_{j=1}^{n} I_j = (0)$.

1.5 Hilbert's Basis Theorem

Definition 1.32. A ring R is called *Noetherian* if every ascending chain of ideals

$$I_1 \subseteq I_2 \subseteq \dots \subseteq I_k \subseteq \dots$$

is stationary, i.e., there exists $N \in \mathbb{N}_{>0}$ such that $I_k = I_N$ for $k \geq N$.

Example 1.33. Any principal ideal domain (PID) is a Noetherian ring, as shown in the course "Algebra".

The following well-known theorem allows us to construct new Noetherian rings from a given one.

Theorem 1.34 (Hilbert's Basis Theorem)**.** *Let R be a Noetherien ring. Then $R[x_1, \dots, x_n]$ is also Noetherian.*

Proof. First of all, it suffices to show the theorem when $n = 1$ since

$$R[x_1, \dots, x_n] = R[x_1, \dots, x_{n-1}][x_n].$$

By contradiction, assume $R[x]$ is not Noetherian. Let $I_1 \subsetneq I_2 \subsetneq \dots$ be a strictly ascending chain of ideals. Set $I := \bigcup_{n \in \mathbb{N}} I_n$. We construct a sequence $\{f_1, f_2, \dots\}$ of polynomials in $R[x]$ as follows:

(a) We choose $f_1 \in I - \{0\}$ of minimal degree.

(b) Assume we found $f_1, f_2, \dots, f_{n-1}$. We pick f_n in $I \setminus (f_1, \dots, f_{n-1})$ of minimal degree.

Note that this is a well-defined construction: indeed, $\mathbb{N}$ is well-ordered and step (b) is repeated infinitely many times since the given chain of ideals is strictly ascending.

Claim 1. $\deg(f_n) \geq \deg(f_{n-1})$ for $n \geq 2$.

Proof of Claim 1. First of all, $\deg(f_2) \geq \deg(f_1)$ by (a). Assume by contradiction $\deg(f_n) < \deg(f_{n-1})$ for some $n \geq 3$. Since by construction $f_n \in I \setminus (f_1, \ldots, f_{n-1}) \subseteq I \setminus (f_1, \ldots, f_{n-2})$, we found a polynomial in $I \setminus (f_1, \ldots, f_{n-2})$ whose degree is lower than $\deg(f_{n-1})$, against the choice of f_{n-1} as a polynomial in $I \setminus (f_1, \ldots, f_{n-2})$ of minimal degree. $\square$

For $i \in \mathbb{N}$ define $a_i \in R$ to be the leading coefficient of f_i and define the ideal $J_i := (a_1, \ldots, a_i)$. We have then an ascending chain of ideals of R, namely $J_1 \subseteq J_2 \subseteq J_3 \subseteq \ldots$. As R is noetherian, there is $N \in \mathbb{N}$ such that $J_k = J_N$ for all $k \geq N$. Consider the element $a_{N+1} \in J_{N+1} = J_N = (a_1, \ldots, a_n)$. Then for $1 \leq i \leq N$ there are $u_i \in R$ such that $a_{N+1} = \sum_{i=1}^{N} u_i a_i$. Now set

$$f'_{N+1} := u_1 x^{\deg(f_{N+1}) - \deg(f_1)} f_1 + \cdots + u_N x^{\deg(f_{N+1}) - \deg(f_N)} f_N.$$

The leading term of f'_{N+1} is $a_{N+1} x^{\deg(f_{N+1})}$ and $f'_{N+1} \in (f_1, \ldots, f_N)$. Consider the polynomial

$$g := f'_{N+1} - f_{N+1}.$$

It holds $g \in I \setminus (f_1, \ldots, f_N)$ since otherwise $f_{N+1} = f'_{N+1} - g \in (f_1, \ldots, f_N)$. Moreover, by construction $\deg(g) < \deg(f_{N+1})$ and this contradicts the choice of f_{N+1} as a polynomial in $I \setminus (f_1, \ldots, f_N)$ of minimal degree. $\square$

Chapter 2

Modules

Definition 2.1. Let A be a ring. An A-*module* is an abelian group $(M, +)$ together with a map $\mu\colon A \times M \to M$, $(a, m) \mapsto a \cdot m := \mu(a, m)$, which satisfies the following properties for all $a, b \in A$ and for all $x, y \in M$:

(a) $a(x + y) = ax + ay$,

(b) $(a + b)x = ax + bx$,

(c) $(ab)x = a(bx)$,

(d) $1_A\, x = x$.

Example 2.2. Let A be a ring. The following are some examples of modules:

(a) An ideal $I \subseteq A$ is an A-module. In particular, A is itself an A-module.

(b) If A is a field, then an A-module is an A-vector space.

(c) Let $(G, +)$ be an abelian group. Then G is a $\mathbb{Z}$-module with respect to the scalar multiplication

$$n \cdot x := \begin{cases} \underbrace{x + \ldots + x}_{n-\text{times}} & \text{if } n \geq 0 \\ \underbrace{(-x) + \ldots + (-x)}_{(-n)-\text{times}} & \text{if } n < 0 \end{cases}$$

(d) Assume $A = k[X]$, where k is a field. Let V be a k-vector space and let $T \in \mathrm{End}(V)$ be a linear map. Then we can regard V as an A-module via the following construction: for $f = \sum_{i=0}^{n} a_i X^i \in k[X]$ set $f(T) := \sum_{i=0}^{n} a_i T^i$ and define $\mu\colon k[X] \times V \to V$ for $f \in k[X]$ and $v \in V$ by

$$f \cdot v := f(T)(v).$$

Definition 2.3. Let A be a ring and M, N be A-modules. A map $f\colon M \to N$ is called an *A-module homomorphism* (or *A-linear*) if the following properties are satisfied:

(i) $f(x+y) = f(x) + f(y)$ for all $x, y \in M$,

(ii) $f(ax) = af(x)$ for all $x \in M$, for all $a \in A$.

Example 2.4. If A is a field, then an A-module homomorphism is a linear map.

Remark 2.5.

(a) Let M, N be A-modules. The set $\mathrm{Hom}_A(M, N)$ of all A-module homomorphisms from M to N is an A-module in the following way. We define the sum of $f, g \in \mathrm{Hom}_A(M, N)$ via

$$(f+g)(x) := f(x) + g(x) \quad \text{for all } x \in M,$$

and the scalar multiplication of $a \in A$ and $f \in \mathrm{Hom}_A(M, N)$ via

$$(af)(x) := af(x) \quad \text{for all } x \in M.$$

These operations make $\mathrm{Hom}_A(M, N)$ into an A-module. When the ring A is clear from the context, we simply write $\mathrm{Hom}(M, N)$.

(b) The composition of A-module homomorphisms is again an A-module homomorphism.

(c) Let M, M', N, N'' be A-modules and let $U\colon M' \to M$ and $V\colon N \to N''$ be A-module homomorphisms. Then we have the following induced A-module homomorphisms:

$$\begin{aligned} \overline{U}\colon\ & \mathrm{Hom}_A(M, N) \longrightarrow \mathrm{Hom}_A(M', N), \quad \overline{U}(f) := f \circ U, \\ \overline{V}\colon\ & \mathrm{Hom}_A(M, N) \longrightarrow \mathrm{Hom}_A(M, N''), \quad \overline{V}(f) := V \circ f. \end{aligned}$$

(d) For any A-module M we have a natural isomorphism

$$\mathrm{Hom}_A(A, M) \cong M, \qquad f \longmapsto f(1).$$

2.1 Submodules and Quotient Modules

Definition 2.6. Let A be a ring and M be an A-module. A *submodule* $N \subseteq M$ is a subgroup of M which is closed under scalar multiplication, i.e., $a \cdot x \in N$ for $a \in A$ and $x \in N$.

Definition 2.7. Let M be an A-module and let $N \subseteq M$ be a submodule. The abelian group M/N inherits an A-module structure from M via

$$a \cdot (x + N) := ax + N \quad \text{for } a \in A,\ x \in M.$$

The A-module M/N is called the *quotient of M by N*. The canonical projection $\pi\colon M \to M/N$ is an A-module homomorphism.

Proposition 2.8. *Let M be an A-module and let $N \subseteq M$ be a submodule. There is a one-to-one inclusion-preserving correspondence between the submodules of M which contain N and submodules of the quotient M/N.*

Definition 2.9. Let A be a ring and let M, N be A-modules. Let $f\colon M \to N$ be an A-module homomorphism.

(a) The *kernel of f* is defined as $\ker(f) := \{x \in M \mid f(x) = 0\}$. This is indeed a submodule of M.

(b) The *image of f* is defined as $\mathrm{im}(f) := \{f(x) \mid x \in M\} = f(M)$. This is indeed a submodule of N.

(c) The *cokernel of f* is defined as the quotient $\mathrm{coker}(f) := N/\mathrm{im}(f)$.

Theorem 2.10 (Homomorphism Theorem/First Isomorphism Theorem)**.** *Let A be a ring, let M, N be A-modules and let $f\colon M \to N$ be an A-module homomorphism. Then*

$$M/\ker(f) \cong \mathrm{im}(f).$$

2.2 Operations on Submodules

Definition 2.11. Let M be an A-module and let $\{M_i\}_{i \in I}$ be a family of submodules of M. We define the *intersection of the submodules M_i*

$$\bigcap_{i \in I} M_i := \Big\{ m \;\Big|\; m \in M_i \text{ for all } i \in I \Big\}$$

and the *sum of the submodules M_i*

$$\sum_{i \in I} M_i := \left\{ \sum_{i \in I} m_i \;\middle|\; m_i \in M_i \text{ for all } i \in I \text{ and } m_i \neq 0 \text{ for only finitely many } i \right\}.$$

Both are submodules of M. Moreover, we say that the sum $\sum_{i \in I} M_i$ is *direct* and denote it by $\bigoplus_{i \in I} M_i$ if every element $m \in \sum_{i \in I} M_i$ can be written uniquely as $\sum_{i \in I} m_i$ where $m_i \in M_i$ for each $i \in I$.

Remark 2.12. Let M_1 and M_2 be submodules of M. The sum $M_1 + M_2$ is direct if and only if $M_1 \cap M_2 = \{0\}$.

Now, assume $\{M_i\}_{i\in I}$ is a chain of submodules of M, i.e., for any $i, j \in I$, either $M_i \subseteq M_j$ or $M_j \subseteq M_i$. Then

$$\bigcup_{i\in I} M_i := \Big\{ m \;\Big|\; m \in M_i \text{ for some } i \in I \Big\}$$

is also a submodule of M.

Since we can't multiply two elements of a module, there is no product of submodules. However, we define the product of an ideal $I \subseteq A$ with the A-module M as

$$IM := \Big\{ \sum_{i=1}^{n} a_i m_i \;\Big|\; a_i \in I,\ m_i \in M,\ n \in \mathbb{N} \Big\},$$

which is a submodule of M.

Theorem 2.13 (Second Isomorphism Theorem)**.** *Let A be a ring, let M be an A-module and let M_1 and M_2 be submodules of M. Then*

$$(M_1 + M_2)/M_1 \cong M_2/(M_1 \cap M_2).$$

Proof. Consider

$$M_2 \longrightarrow M_1 + M_2 \longrightarrow (M_1 + M_2)/M_1, \qquad x \longmapsto 0 + x \longmapsto x + M_1.$$

The composition $\varphi\colon M_2 \to (M_1 + M_2)/M_1$ is surjective since for $m_i \in M_i$ it holds $(m_1 + m_2) + M_1 = m_2 + M_1$. Moreover, we have $\ker(\varphi) = M_1 \cap M_2$. Now the first isomorphism theorem gives the thesis. □

Theorem 2.14 (Third Isomorphism Theorem)**.** *Let A be a ring. Let $L \supseteq M \supseteq N$ be A-modules. Then*

$$(L/N)/(M/N) \cong L/M.$$

Proof. Consider

$$\psi\colon L/N \longrightarrow L/M, \qquad x + N \longmapsto x + M.$$

This is a well-defined A-module homomorphism since $N \subseteq M$. It is clearly surjective and moreover $\ker(\psi) = M/N$. Therefore, the first isomorphism theorem shows the assertion. □

2.3 Direct Sums and Direct Products

Definition 2.15. Let A be a ring and let $\{M_i\}_{i\in I}$ be a family of A-modules. We define:

(i) The *direct product of the* M_i

$$\prod_{i\in I} M_i := \Big\{ f\colon I \to \bigsqcup_{i\in I} M_i \;\Big|\; f(i) \in M_i \text{ for } i \in I \Big\}.$$

This is an A-module whose elements are usually written as sequences $(m_i)_{i\in I}$ such that $m_i \in M_i$ for $i \in I$. The operations in $\prod_{i\in I} M_i$ are componentwise.

(ii) The *direct sum of the* M_i

$$\bigoplus_{i\in I} M_i := \Big\{ f \in \prod_{i\in I} M_i \;\Big|\; f(i) \neq 0 \text{ for only finitely many } i \Big\}.$$

This is a submodule of the direct product $\prod_{i\in I} M_i$.

Remark 2.16. In general, direct sum and direct product of a family $\{M_i\}_{i\in I}$ of A-modules are not the same. They coincide if the index set I is finite.

Remark 2.17 (Internal and External Direct Sum)**.** Let M be an A-module. For a family $\{M_i\}_{i\in I}$ of submodules of M we have at this stage two notions of their direct sum: namely, the direct sum as submodules in the sense of Definition 2.11 and the direct sum as a family of modules in the sense of Definition 2.15. For the time being, we refer to them, respectively, as *internal direct sum* and *external direct sum*. Indeed, the two concepts coincide up to isomorphism and this is the reason why one refers to both of them simply as direct sum. For the benefit of the reader, we explain in detail the case of two submodules M_1 and M_2. Suppose that the sum $M_1 + M_2$ is direct in the sense of Definition 2.11 and denote it by $M_1 \oplus_i M_2$. On the other hand, we consider the external direct sum of M_1 and M_2, which we denote here by $M_1 \oplus_e M_2$. Consider

$$f\colon M_1 \oplus_i M_2 \to M_1 \oplus_e M_2, \quad m_1 + m_2 \mapsto (m_1, m_2).$$

This map is well-defined since the sum $M_1 + M_2$ is direct in the sense of Definition 2.11. It is easy to see that f is indeed an isomorphism of A-modules.

Let now M_1 and M_2 be A-modules. Consider the external direct sum $M_1 \oplus_e M_2$. We define:

$$\widetilde{M}_1 := \{(m_1, 0) \mid m_1 \in M_1\} \qquad \text{and} \qquad \widetilde{M}_2 := \{(0, m_2) \mid m_2 \in M_2\}.$$

Both are submodules of $M_1 \oplus_e M_2$ and clearly $\widetilde{M}_i \cong M_i$. It is easy to see that the sum $\widetilde{M}_1 + \widetilde{M}_2$ is direct in the sense of Definition 2.11. Moreover, we have

$$\widetilde{M}_1 \oplus_i \widetilde{M}_2 = M_1 \oplus_e M_2.$$

Remark 2.18 (Direct Sum of Rings and Direct Sum of Ideals)**.**

Let A be a nontrivial ring.

(i) Assume that A is a direct sum of rings $A = \bigoplus_{j=1}^n A_j = A_1 \times \ldots \times A_n$. Then the ring

$$I_j := \{(0, \ldots, 0, a_j, 0, \ldots, 0) \mid a_j \in A_j\} \cong A_j$$

is an ideal of A, but not a subring since $1_A = (1, \ldots, 1) \notin I_j$. Hence, the ring A, considered as an A-module, is the direct sum of the ideals I_j, i.e., $A = \bigoplus_{j=1}^n I_j$.

(ii) Assume now that A, considered as an A-module, is a direct sum of ideals $A = \bigoplus_{j=1}^n I_j$. Set $B_j := \bigoplus_{k \neq j} I_k$ and note that $A/B_j \cong I_j$, which shows that I_j is indeed a ring, and hence there is a ring isomorphism

$$A \cong \bigoplus_{j=1}^n (A/B_j).$$

Since $A = \bigoplus_{j=1}^n I_j$, we may write $1_A = (y_1, \ldots, y_n)$ with $y_j \in I_j$. Then we have that

$$1_{A/B_j} = 1_A + B_j = (y_1, \ldots, y_n) + B_j = (0, \ldots, 0, y_j, 0, \ldots, 0) + B_j$$

and by means of the isomorphism $A/B_j \cong I_j$ we get $1_{I_j} = y_j$. Note that $1_{I_j} = y_j$ is idempotent and $I_j = (y_j)$ as an ideal in A.

2.4 Finitely Generated Modules

We now introduce some terminology and key concepts needed for what follows.

Definition 2.19. Let A be a ring and M be an A-module.

(a) Let $x \in M$. The set $Ax := \{ax \mid a \in A\} \subseteq M$ is a submodule of M which is called *the submodule generated by* x.

(b) The A-module M is called *principal* if there exists $x \in M$ such that $M = Ax$.

(c) Let $\{x_i\}_{i \in I}$ be a subset of M. The sum $\sum_{i \in I} Ax_i$ is called *the submodule generated by the subset* $\{x_i\}_{i \in I}$.

(d) If $M = \sum_{i \in I} Ax_i$, then the set $\{x_i\}_{i \in I}$ is called a *set of generators of* M.

(e) The A-module M is called *finitely generated* if there exists a finite set of generators of M, i.e., $M = Ax_1 + \ldots + Ax_n$ for some $x_1, \ldots, x_n \in M$.

(f) The A-module M is called *free* if $M \cong \bigoplus_{i \in I} M_i$, where $M_i \cong A$ for each $i \in I$.

(g) For $n \in \mathbb{N}$, we denote by A^n the direct sum of A with itself n times:

$$A^n := \bigoplus_{i=1}^{n} A = \underbrace{A \oplus \ldots \oplus A}_{n-\text{times}}.$$

For $i \in \{1, \ldots, n\}$, we define $e_i := (0, \ldots, 0, \overset{i}{1}, 0, \ldots, 0)$. The set $\{e_i\}_{i=1}^n$ is called the *canonical basis* (or *canonical set of generators*) of A^n. By convention, we set $A^0 := \{0\}$.

(h) Let X be a set. We define

$$A^X := \Big\{ f \colon X \to A \mid f(x) \neq 0 \text{ for only finitely many } x \Big\}.$$

For $f, g \in A^X$ and for $a \in A$ we define for each $x \in X$

$$(f+g)(x) := f(x) + g(x) \quad \text{and} \quad (a \cdot f)(x) := af(x).$$

The operations above make A^X into an A-module. Indeed, there is an A-linear isomorphism $A^X \cong \bigoplus_{x \in X} M_x$, where $M_x = A$ for all $x \in X$. Thus, A^X is a free module called *the free module over* X. For each $x \in X$ define $e_x \in A^X$ by

$$e_x(y) = \delta_{xy} = \begin{cases} 1 & \text{if } x = y, \\ 0 & \text{if } x \neq y. \end{cases}$$

The set $\{e_x\}_{x \in X}$ is called the *canonical basis* (or *canonical set of generators*) of A^X.

Remark 2.20. (a) Note that the free module A^X can be described as the A-module S of finite formal linear combinations

$$\sum_{i=1}^{n} a_i x_i,$$

where $n \in \mathbb{N}_{>0}$, $a_i \in A$ and $x_i \in X$. Indeed, there is an A-linear isomorphism

$$A^X \to S, \quad f \mapsto \sum_{x \in X} f(x) \cdot x,$$

where the sum is finite since $f(x) \neq 0$ for only finitely many $x \in X$. We will adopt the description of A^X as S, since it is more convenient for computations and constructions. Note that in this description of A^X the canonical basis is given by the set X itself: each $x \in X$ can be regarded as the formal linear combination $1 \cdot x$.

(b) Note that the trivial module $M = 0$ is finitely generated: the unique set of generators is the empty-set.

We will mostly focus on finitely generated modules. The following result provides a useful characterization of these modules.

Proposition 2.21. *Let A be a ring and let M be an A-module. Then M is a finitely generated A-module if and only if it is isomorphic to a quotient of a finitely generated free A-module. More precisely, $M \cong A^n/B$ if and only if M has n generators.*

Proof. "$\Rightarrow$": Let $x_1, \ldots, x_n$ be generators of M. Define

$$\varphi\colon A^n \longrightarrow M, \qquad (a_1, \ldots, a_n) \longmapsto \sum_{i=1}^{n} a_i x_i.$$

The map φ is a surjective A-module homomorphism, and thus $A^n/\ker(\varphi) \cong M$ by the homomorphism theorem.

"$\Leftarrow$": If $M \cong A^n/B$, we have a surjective A-module homomorphism $\varphi\colon A^n \to M$. Then the images $\varphi(e_i)$ of the canonical generators e_i generate M. □

Proposition 2.22 (Nakayama's Lemma)**.** *Let A be a ring and let M be a finitely generated A module. Let I be an ideal of A contained in the Jacobson radical J_A. Then $IM = M$ implies $M = \{0\}$.*

Proof. We argue by contradiction. Suppose $M \neq \{0\}$ and let $u_1, \ldots, u_n$ be a minimal set of generators of M. Then $u_n \in M = IM$, thus we may write $u_n = a_1 u_1 + \cdots + a_n u_n$ for some $a_i \in I$. But then

$$(1 - a_n)u_n = a_1 u_1 + \cdots + a_{n-1} u_{n-1} \tag{2.1}$$

and since $a_n \in I \subseteq J_A$, it follows from Proposition 1.26 that $1 - a_n$ is a unit in A. Hence, by multiplying (2.1) with $(1 - a_n)^{-1}$ we have that $u_n \in Au_1 + \ldots + Au_{n-1}$, a contradiction since the set of u_i is a minimal set of generators of M. □

Corollary 2.23. *Let A be a ring and let M be a finitely generated A-module. Let $N \subseteq M$ be a submodule and let I be an ideal contained in the Jacobson radical J_A. Then $M = IM + N$ implies $N = M$.*

Proof. Observe that

$$I(M/N) = (IM + N)/N.$$

Since $M = IM + N$, we have

$$I(M/N) = M/N.$$

Moreover, because M is is finitely generated, the quotient module M/N is finitely generated as well. Therefore, we apply Nakayama's Lemma to M/N and get $M/N = 0$, equivalently, $N = M$. □

Definition 2.24. Let A be a ring and let M be an A-module. The *annihilator of* M is defined as

$$\mathrm{Ann}(M) := \{a \in A \mid am = 0 \text{ for all } m \in M\}$$

This is an ideal in A.

Definition 2.25. An A-module M is called *faithful* if $\mathrm{Ann}(M) = 0$.

Remark 2.26. (a) Let M be an A-module and let $I \subseteq \mathrm{Ann}(M)$ be an ideal. Then we may regard M as an (A/I)-module as follows: for $a \in A$ and for $x \in M$, we define

$$(a + I) \cdot x := ax.$$

This scalar multiplication is well-defined since $IM = 0$.

(b) Let M be a finitely generated A-module and let $\{x_1, \ldots, x_n\}$ be a minimal set of generators of M. Let $I \subseteq J_A$ be an ideal. Since $I \subseteq \mathrm{Ann}(M/IM)$, the quotient M/IM has a natural structure of (A/I)-module by (a). Denote by $[x_1], \ldots, [x_n]$ the residue classes of $x_1, \ldots, x_n$ in M/IM; these classes generate M/IM as an A-module as well as an (A/I)-module. Since the x_i's form a minimal set of generators, by using the same argument as in the proof of Nakayama's lemma we have that $[x_j] \neq 0$ for $1 \leq j \leq n$ and $[x_i] \neq [x_j]$ for any $i \neq j$.

Let $(A, \mathfrak{m}, k)$ be a local ring and let M be a finitely generated A-module. Since $\mathfrak{m} \subseteq \mathrm{Ann}(M/\mathfrak{m}M)$, by Remark 2.26(a) the quotient module $M/\mathfrak{m}M$ can be regarded as an $(A/\mathfrak{m})$-module, i.e., a k-vector space. Since M is a finitely generated A-module, then $M/\mathfrak{m}M$ is a finitely generated A-module, hence a finite dimensional k-vector space. Let $\{x_1, \ldots, x_n\}$ be a minimal set of generators of M and denote by $[x_1], \ldots, [x_n]$ the residue classes of $x_1, \ldots, x_n$ in $M/\mathfrak{m}M$. By Remark 2.26(b) the classes $[x_1], \ldots, [x_n]$ are pairwise distinct. Indeed, we can say more.

Proposition 2.27. *Let $(A, \mathfrak{m}, k)$ be a local ring and let M be a finitely generated A-module. Let $\{x_1, \ldots, x_n\}$ be a minimal set of generators of M. Then the classes $[x_1], \ldots, [x_n]$ form a basis of $M/\mathfrak{m}M$ as a k-vector space. In particular, any minimal set of generators of M contains exactly* $\dim_k(M/\mathfrak{m}M)$ *elements.*

Proof. It is enough to show that $[x_1], \ldots, [x_n]$ are linearly independent over k.

Let

$$\overline{a_1}[x_1] + \ldots + \overline{a_n}[x_n] = [0], \quad \overline{a_i} \in A/\mathfrak{m} = k.$$

Hence, for $1 \leq i \leq n$ there exists $b_i \in \mathfrak{m}$ such that

$$a_1x_1 + \ldots + a_nx_n = b_1x_1 + \ldots + b_nx_n \in \mathfrak{m}M.$$

Assume by contradiction that $\overline{a_i} \neq \overline{0}$, i.e., $a_i \notin \mathfrak{m}$, for some i. Then $a_i - b_i \notin \mathfrak{m}$ since $b_i \in \mathfrak{m}$, and hence $a_i - b_i$ is invertible since $\mathfrak{m}$ is the unique maximal ideal of the local ring A. It follows that

$$x_i = \sum_{j \neq i} (a_i - b_i)^{-1}(b_j - a_j)x_j,$$

against the minimality of $\{x_1, \dots, x_n\}$. □

Hence, if $\{x_1, \dots, x_n\}$ is a minimal set of generators of M, then $\{[x_1], \dots, [x_n]\}$ is a basis of the k-vector space $M/\mathfrak{m}M$. The following result shows that the converse holds true as well.

Proposition 2.28 (Nakayama's Lemma for Local Rings)**.** *Let $(A, \mathfrak{m}, k)$ be a local ring and let M be a finitely generated A-module. Let $x_1, \dots, x_n$ be elements of M whose residue classes $[x_1], \dots, [x_n] \in M/\mathfrak{m}M$ form a basis of this k-vector space. Then $x_1, \dots, x_n$ generate M.*

Proof. If $M/\mathfrak{m}M = 0$, then the unique subset of M fulfilling the assumptions is the empty-set. The thesis is then true since $M = 0$ by Nakayama's lemma. Now assume $M/\mathfrak{m}M \neq 0$. Then there are elements $x_1, \dots, x_n \in M$ fulfilling the assumptions. Let $N := \sum_{i=1}^n Ax_i$. Observe that the map A-linear map

$$\varphi\colon N \to M/\mathfrak{m}M, \qquad x \mapsto [x]$$

is surjective since $\{[x_i] \mid 1 \leq i \leq n\}$ is a basis of $M/\mathfrak{m}M$.

Since M is finitely generated and $J_A = \mathfrak{m}$, if we show that $N + \mathfrak{m}M = M$, then $N = M$ by Corollary 2.23 and we are done. Hence, let us show that $N + \mathfrak{m}M = M$.

"$\subseteq$" is clear. For "$\supseteq$", let $x \in M$ and consider $[x] \in M/\mathfrak{m}M$. Since φ is surjective, there is $n \in N$ such that $[n] = [x]$, and thus $x - n \in \mathfrak{m}M$. Therefore, there exists $y \in \mathfrak{m}M$ such that $x = n + y \in N + \mathfrak{m}M$. □

Remark 2.29. Let $(A, \mathfrak{m}, k)$ be a local ring. Then the following statements are equivalent:

(a) Let M be a finitely generated A-module. If $M = \mathfrak{m}M$, then $M = 0$.

(b) Let M be a finitely generated A-module. Let $x_1, \dots, x_n$ be elements of M whose residue classes $[x_1], \dots, [x_n] \in M/\mathfrak{m}M$ form a basis of this k-vector space. Then $x_1, \dots, x_n$ generate M.

Hence, for a local ring Nakayma's Lemma (Proposition 2.22) and Proposition 2.28 are indeed equivalent and this motivates the name "Nakayama's lemma for local rings" given to Proposition 2.28.

2.5 Exact Sequences

Definition 2.30. Let A be a ring. A sequence of A-modules and A-module homomorphisms

$$\cdots \longrightarrow M_{i-1} \xrightarrow{f_i} M_i \xrightarrow{f_{i+1}} M_{i+1} \longrightarrow \dots \tag{2.2}$$

is called

(i) *exact at* M_i if $\ker(f_{i+1}) = \operatorname{im}(f_i)$,

(ii) *exact* if it is exact at every M_i.

In particular, a sequence $0 \to M' \xrightarrow{f} M$ is exact if and only if f is injective, a sequence $M \xrightarrow{g} M'' \to 0$ is exact if and only if g is surjective and

$$0 \longrightarrow M' \xrightarrow{f} M \xrightarrow{g} M'' \longrightarrow 0$$

is exact if and only if f is injective, g is surjective and $\ker(g) = \operatorname{im}(f)$. A sequence of this form is called a *short exact sequence*.

Remark 2.31. Any long exact sequence of the form 2.2 can be split into short exact sequences. Indeed, for each i set $N_i := \operatorname{im}(f_i) = \ker(f_{i+1})$. Then we have a short exact sequence

$$0 \longrightarrow N_i \xrightarrow{j} M_i \xrightarrow{f_{i+1}} N_{i+1} \longrightarrow 0,$$

where $j\colon N_i \to M_i$ is the inclusion map.

Proposition 2.32. *(i) Let*

$$M' \xrightarrow{u} M \xrightarrow{v} M'' \longrightarrow 0 \tag{2.3}$$

be a sequence of A-modules and A-module homomorphisms. Then (2.3) *is exact if and only if for all A-modules N the sequence*

$$\begin{array}{ccccc} 0 \longrightarrow \operatorname{Hom}_A(M'', N) & \xrightarrow{\overline{v}} & \operatorname{Hom}_A(M, N) & \xrightarrow{\overline{u}} & \operatorname{Hom}_A(M', N) \\ f \longmapsto f \circ v & & & & \\ & & & & g \longmapsto g \circ u \end{array}$$

is exact.

(ii) Let

$$0 \longrightarrow N' \xrightarrow{u} N \xrightarrow{v} N'' \tag{2.4}$$

be a sequence of A-modules and A-module homomorphisms. Then (2.4) *is exact if and only if for all A-modules M the sequence*

$$\begin{array}{ccccc} 0 \longrightarrow \operatorname{Hom}_A(M, N') & \xrightarrow{\overline{u}} & \operatorname{Hom}_A(M, N) & \xrightarrow{\overline{v}} & \operatorname{Hom}_A(M, N'') \\ f \longmapsto u \circ f & & & & \\ & & & & g \longmapsto v \circ g \end{array}$$

is exact.

Proposition 2.33 (Snake Lemma)**.** *Let*

$$\begin{tikzcd}
 & M' \arrow[r, "u"] \arrow[d, "f'"] & M \arrow[r, "v"] \arrow[d, "f"] & M'' \arrow[r] \arrow[d, "f''"] & 0 \\
0 \arrow[r] & N' \arrow[r, "u'"] & N \arrow[r, "v'"] & N''
\end{tikzcd}$$

be a commutative diagram of A-modules and A-module homomorphisms with exact rows. Then there is an exact sequence

$$\begin{tikzcd}
 & \ker(f') \arrow[r, "\overline{u}"] & \ker(f) \arrow[r, "\overline{v}"] & \ker(f'') \arrow[dll, "d"] \\
 & \operatorname{coker}(f') \arrow[r, "\overline{u}'"] & \operatorname{coker}(f) \arrow[r, "\overline{v}'"] & \operatorname{coker}(f'')
\end{tikzcd} \tag{2.5}$$

where $\overline{u}$, $\overline{v}$ are restrictions of u, v and $\overline{u}'$, $\overline{v}'$ are induced by u', v'. The map d is called the connecting homomorphism *(or* boundary homomorphism*). Moreover, if u is injective, then so is the first map in* (2.5)*, and if v' is surjective, then so is the last map in* (2.5)*.*

Proof. We have a diagram as follows:

$$\begin{tikzcd}
 & 0 \arrow[d] & 0 \arrow[d] & 0 \arrow[d] & \\
0 \arrow[r, dotted] & \ker(f') \arrow[r, dotted, "\overline{u}"] \arrow[d] & \ker(f) \arrow[r, dotted, "\overline{v}"] \arrow[d] & \ker(f'') \arrow[r, dotted] \arrow[d] & 0 \arrow[dddll, dotted, "d"] \\
0 \arrow[r, dotted] & M' \arrow[r, "u"] \arrow[d, "f'"] & M \arrow[r, "v"] \arrow[d, "f"] & M'' \arrow[r] \arrow[d, "f''"] & 0 \\
0 \arrow[r] & N' \arrow[r, "u'"] \arrow[d] & N \arrow[r, "v'"] \arrow[d] & N' \arrow[r, dotted] \arrow[d] & 0 \\
 & \operatorname{coker}(f') \arrow[r, dotted, "\overline{u}'"] \arrow[d] & \operatorname{coker}(f) \arrow[r, dotted, "\overline{v}'"] \arrow[d] & \operatorname{coker}(f'') \arrow[r] \arrow[d] & 0 \\
 & 0 & 0 & 0 &
\end{tikzcd}$$

Define $\overline{u} := u_{|\ker(f')}\colon \ker(f') \to \ker(f)$. We have to check just that $f(u(x)) = 0$ for $x \in \ker(f')$. Since the diagram above is commutative, we have $f(u(x)) = u'(f'(x)) = u'(0) = 0$. Similarly, one can check that $\overline{v} := v_{|\ker(f)}\colon \ker(f) \to \ker(f'')$ is well-defined. Note that $\overline{u}'$ (resp. $\overline{v}'$) is well-defined if $u'(\operatorname{im}(f')) \subseteq \operatorname{im}(f)$ (resp. $v'(\operatorname{im}(f)) \subseteq \operatorname{im}(f'')$). Let $y = f'(x) \in \operatorname{im}(f')$. Then $u'(y) = u'(f'(x)) = f(u(x))$, hence $u'(y) \in \operatorname{im}(f)$. Similarly for $v'(\operatorname{im}(f)) \subseteq \operatorname{im}(f'')$.

Define now $d\colon \ker(f'') \to \operatorname{coker}(f')$ as follows. Let $x \in \ker(f'') \subseteq M''$. Since v is surjective, there exists $t \in M$ such that $v(t) = x$. Hence, $0 = f''(x) = f''(v(t)) = v'(f(t))$, and thus $f(t) \in \ker(v') = \operatorname{im}(u')$, i.e., there exists $y \in N'$ such that

$u'(y) = f(t)$ and we can define $d(x) := [y] = y + \operatorname{im}(f')$. For the time being, it seems that our definition of d depends on two choices: the choice of $t \in M$ such that $v(t) = x$ and the choice of $y \in N'$ such that $u'(y) = f(t)$. Hence, let $t' \in M$ such that $v(t') = x = v(t)$ and let $y' \in N'$ such that $u'(y') = f(t')$. We have to show that $y + \operatorname{im}(f') = y' + \operatorname{im}(f')$, i.e., $y - y' \in \operatorname{im}(f')$. Observe that by definition of t and t' we have $t - t' \in \ker(v) = \operatorname{im}(u)$, i.e., there exists $z \in M'$ such that $u(z) = t - t'$. On the other hand, by definiton of y and y' we have

$$u'(y - y') = f(t - t') = f(u(z)) = u'(f'(z)),$$

hence

$$y - y' - f'(z) \in \ker(u') = 0 \implies y - y' = f'(z).$$

Finally, $y - y' \in \operatorname{im}(f')$ and this shows that the map d is well-defined. It is now easy to check that d is an A-module homomorphism and this is left to the reader. As for the exactness of (2.5), we observe for example that

$$\operatorname{im}(\overline{u}) = \operatorname{im}(u)_{|\ker(f)} = \ker(v)_{|\ker(f)} = \ker(\overline{v}),$$

where the first equality holds because $f \circ u = u' \circ f'$ and u' is injective, the second equality is true because $\operatorname{im}(u) = \ker(v)$ and the third equality is immediate since $\overline{v}$ is the restriction of v to $\ker(f)$. By diagram chasing the reader can fully check the exactness of (2.5). □

Definition 2.34. Let $\mathcal{C}$ be a class of A-modules. A function $\lambda\colon \mathcal{C} \to \mathbb{Z}$ is called *additive* if for every short exact sequence $0 \to M' \to M \to M'' \to 0$ it holds that $\lambda(M') - \lambda(M) + \lambda(M'') = 0$.

Example 2.35. Let $A = k$ be a field and $\mathcal{C}$ be the class of finite-dimensional k-vector spaces. Then $\lambda\colon \mathcal{C} \to \mathbb{Z}$, $V \to \dim_k V$, is additive.

Proposition 2.36. *Let*

$$0 \longrightarrow M_0 \longrightarrow M_1 \longrightarrow \cdots \longrightarrow M_n \longrightarrow 0$$

be an exact sequence of A-modules where all the modules M_i and the kernels of all homomorphisms belong to $\mathcal{C}$. Then for any additive function $\lambda\colon \mathcal{C} \to \mathbb{Z}$ on $\mathcal{C}$ we have

$$\sum_{i=0}^{n} (-1)^i \lambda(M_i) = 0.$$

Proof. Like in Remark 2.31, split the exact sequence into short exact sequences $0 \to N_i \to M_i \to N_{i+1} \to 0$, where $N_0 = N_{n+1} = 0$. Then for each $0 \le i \le n$ we have $\lambda(M_i) = \lambda(N_i) + \lambda(N_{i+1})$, namely:

$$\begin{aligned}
\lambda(M_0) &= \lambda(N_0) + \lambda(N_1) \\
\lambda(M_1) &= \lambda(N_1) + \lambda(N_2) \\
&\cdots \\
\lambda(M_{n-1}) &= \lambda(N_{n-1}) + \lambda(N_n) \\
\lambda(M_n) &= \lambda(N_n) + \lambda(N_{n+1})
\end{aligned}$$

We sum these $n+1$ equality with alternating signs, and since $\lambda(N_0) = \lambda(N_{n+1}) = 0$, on the right-hand side all terms cancel pairwise, yielding

$$\sum_{i=0}^{n}(-1)^i\lambda(M_i) = 0.$$

$\square$